

# Pointed convex cones

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A convex cone  $K \subseteq \mathbb{R}^n$  can contain a non-trivial linear set, *i.e.*, a linear set  $L \subseteq K$  with  $L \neq \{0_n\}$ . Observe that  $K$  contains a non-trivial linear set if and only if  $K$  contains a 1-dimensional line. A convex cone that does *not* contain a non-trivial linear set is called a *pointed* convex cone:

**Definition 1** (*Pointed convex cones*)

A convex cone is called *pointed* if it does not contain a line through the origin. ■

The key characteristic of pointed convex cones is that pointed convex cones are exactly those convex cones, where the origin is an extreme point:

**Theorem 1** (*The origin is an extreme point of a pointed convex cone*)

A convex cone  $K \subseteq \mathbb{R}^n$  is pointed  $\iff$  the origin is an extreme point of  $K$ . ■

*Proof:* " $\Rightarrow$ " : Suppose  $K \subseteq \mathbb{R}^n$  is a pointed convex cone. Suppose, for a contradiction, that  $0_n = \sum_{i=1}^k \lambda_i d^i$ , where  $d^1, \dots, d^k \in K \setminus \{0_n\}$ ,  $\lambda > 0_k$  and  $\sum_{i=1}^k \lambda_i = 1$ . Isolating  $d^k$  gives:

$$-d^k = \sum_{i=1}^{k-1} \frac{\lambda_i}{\lambda_k} d^i,$$

which implies  $-d^k, d^k \in K$ . Hence  $\text{cone}(\{-d^k, d^k\}) = \text{span}(\{d^k\}) \subseteq K$ . However, this contradicts that  $K$  is pointed.

" $\Leftarrow$ " : Now suppose  $K$  is *not* pointed. This implies  $\text{span}(\{d\}) \subseteq K$  for some  $d \neq 0_n$ . In particular  $-d, d \in K \setminus \{0_n\}$ . Since  $0_n = \frac{1}{2} \cdot d + \frac{1}{2} \cdot (-d)$ , the origin is a convex combination of points in  $K \setminus \{0_n\}$  and not extreme. ■

Since the origin is a convex combination of non-zero directions in a cone if and only if the origin is a sum of non-zero directions in the cone, one also has the following characterization of pointed convex cones:

**Observation 1** (*Characterization of pointed convex cones*)

A convex cone  $K \subseteq \mathbb{R}^n$  is pointed  $\iff$  the origin is not a sum of directions in  $K \setminus \{0_n\}$ . ■